

Client Change Request (CCR)**CCR Number:** CCR-007**Client:** Green Valley Supply Company**Submitted By:** David Mitchell, Inventory Manager**Priority:** High**Requested Delivery:** End of Today's Sprint**Assigned Developer:** _____**Date Assigned:** _____

Business Background

Green Valley Supply Company maintains an inventory of products that are sold throughout the day. Every time a customer makes a purchase, the inventory must be updated to reflect the items that have been sold.

Currently, employees manually subtract sold items from the inventory and then record the updated inventory count on paper. This process is time-consuming and occasionally results in incorrect inventory totals.

The Inventory Manager would like a simple computer program that keeps track of the current inventory while allowing employees to process one customer purchase at a time. Some information, such as the total inventory available, must remain available throughout the entire program. Other information, such as the number of items sold during an individual customer purchase, is only needed while processing that sale.

At this time, the manager is interested in processing one purchase during each execution of the program.

Customer Requirements

The employee should be able to:

- Enter the current number of products available in inventory.
- Enter the number of products purchased by a customer.

The program should:

- Determine the updated inventory after the sale.
- Display the number of products sold during the current customer purchase.
- Display the updated inventory remaining after the sale.

The inventory remaining should accurately reflect the sale that was just completed.

The report should be easy for employees to read and verify.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

- ☐ Employee can enter the current inventory quantity.
 - ☐ Employee can enter the quantity purchased by the customer.
 - ☐ Program correctly calculates the remaining inventory.
 - ☐ Program displays the quantity sold during the current transaction.
 - ☐ Program displays the updated inventory.
 - ☐ Program produces a clear and professionally formatted report.
 - ☐ Program completes without errors.
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Out of Scope

The customer is **not** requesting:

- Processing multiple customer purchases
- Saving inventory information to a file
- Product descriptions
- Product numbers or SKUs
- Multiple inventory locations
- Inventory reports
- Inventory reordering
- Barcode scanners
- Databases
- Graphical user interfaces (GUI)

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Employee Input

Current Inventory: 250

Products Purchased: 18

Expected Output

Green Valley Supply Company

Inventory Transaction Summary

Inventory Before Sale : 250

Products Sold : 18

Inventory Remaining : 232

Example Scenario 2

Employee Input

Current Inventory: 75

Products Purchased: 20

Expected Output

Green Valley Supply Company

Inventory Transaction Summary

Inventory Before Sale : 75

Products Sold : 20

Inventory Remaining : 55

Example Scenario 3

Employee Input

Current Inventory: 15

Products Purchased: 15

Expected Output

Green Valley Supply Company

Inventory Transaction Summary

Inventory Before Sale : 15

Products Sold : 15

Inventory Remaining : 0

Notes from the Customer

"I don't want employees trying to remember inventory numbers throughout the day. The computer should always know how many products remain after each sale. Information about the current customer purchase is only needed while that sale is being processed, but the overall inventory should always remain available."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

- 1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 6**, including:

- Variables and constants
- Input/output
- Formatted output
- Selection statements (if and switch)
- Loops
- Functions
- Local and global variables (scope)

Do **not** use C language features that have not yet been introduced in the course.